The random Module in Python

Let your Python programs roll dice, pick winners and shuffle cards, all by chance!

The Big Idea

- The random module is a toolbox that comes with Python.
- It helps your program make choices by chance, like rolling a dice.
- You must write import random at the top before you use it.
- Every tool inside it is used by writing random. followed by the tool name.
- Games, quizzes and lucky draws often use it.

Key Tools You Will Use

- random.randint(a, b) gives a whole number from a to b. Both ends can come up.
- random.random() gives a decimal number from 0.0 up to, but never including, 1.0.
- random.choice(list) picks one item from a list.
- random.shuffle(list) mixes up the order of a list.
- random.sample(list, k) picks k different items from a list.
- random.uniform(a, b) gives a decimal number between a and b.

Step-by-Step: Using the Module

- Step 1: Write import random on the first line.
- Step 2: Pick the tool you need, such as random.randint.
- Step 3: Put the values inside the brackets.
- Step 4: Save the answer in a variable or print it.
- Step 5: Run the program again. The answer can change each time!

📋 Lists and the random Module

- A list holds many items inside square brackets, like ['red', 'blue', 'green'].
- random.choice() picks just one item from the list.
- random.sample() picks several different items, with no repeats.
- random.shuffle() changes the list itself. It does not give back a new list.
- Shuffle only works on lists, not on plain text or a tuple.

🔍 Is It Really Random?

- Python's numbers look random, but a math recipe makes them.
- This is called pseudo-random, which means almost random.
- random.seed(number) starts the recipe at the same place every time.
- With the same seed, you get the same results each run.
- Seeds are handy when you want to test or repeat your program.

🌎 Real-Life Connections

- A dice game uses random.randint(1, 6).
- A lucky draw in class uses random.choice() on a list of names.
- A quiz app shuffles the questions so they come in a new order.
- A card game shuffles the deck before dealing.
- A game picks a random spot where a treasure appears.

💻 Code Examples

Roll a Dice python

```python
import random

# Pick a whole number from 1 to 6
roll = random.randint(1, 6)
print("You rolled:", roll)
```

▶ Output:

```
You rolled: 4
```

This program rolls a dice and shows a number from 1 to 6. Run it again to see a new number.

Pick a Winner and Shuffle python

```python
import random

names = ["Asha", "Ben", "Chitra", "Dev"]

# Pick one name
winner = random.choice(names)
print("Winner:", winner)

# Mix up the list
random.shuffle(names)
print("Shuffled:", names)
```

▶ Output:

```
Winner: Chitra
Shuffled: ['Dev', 'Asha', 'Chitra', 'Ben']
```

This uses choice to pick one name and shuffle to mix the list. The names shown will change each time you run it.

Seed Makes It Repeat python

```python
import random

random.seed(1)
first = random.randint(1, 100)

random.seed(1)
second = random.randint(1, 100)

# Both numbers are the same because the seed is the same
print(first == second)
```

▶ Output:

```
True
```

With the same seed, random.randint gives the same numbers each time. Two runs with seed 1 will match each other.

📚 Key Terms

module	A ready-made set of tools that you can add to your Python program.
import	The word that brings a module into your program.
randint	Gives a random whole number between two numbers, including both.
random()	Gives a random decimal number from 0.0 up to but not including 1.0.
choice	Picks one random item from a list.
shuffle	Mixes the items of a list into a new random order.
sample	Picks several different items from a list without repeating.
seed	A starting number that makes the random results repeat the same way.
pseudo-random	Looks random but is made by a fixed math recipe.
list	A group of items kept in square brackets.

🧮 Formulas & Rules

```
import random
```

```
random.randint(a, b) # whole number from a to b, both included
```

```
random.random() # decimal from 0.0 up to but not including 1.0
```

```
random.uniform(a, b) # decimal between a and b
```

```
random.choice(my_list) # one random item
```

```
random.shuffle(my_list) # mixes the list in place
```

```
random.sample(my_list, k) # k different items
```

```
random.seed(number) # makes results repeatable
```

📝 Worked Examples

Q: How do you roll a dice that shows 1 to 6?
A: Step 1: import random. Step 2: use random.randint(1, 6). Step 3: print the answer. Each run shows a number from 1 to 6.

Q: How do you pick one winner from a list of names?
A: Make a list of names. Use random.choice(names) and print the result. One name is picked each time.

Q: How do you pick 3 different students from a list of 5?
A: Use random.sample(students, 3). It returns a list of 3 different names, so nobody is picked twice.

🚫 Common Mistakes

- Forgetting import random, which gives a NameError.
- Writing randint(1, 6) without random. in front of it.
- Thinking randint(1, 6) never gives 6. It can give 6, because both ends are included.
- Writing my_list = random.shuffle(my_list). Shuffle changes the list itself and gives back None.
- Naming your own file random.py. Python then gets confused and your program breaks.

👫 Parent Corner

- Ask your child to explain what import random does, in their own words.
- Roll a real dice 20 times and write down the results. Then run random.randint(1, 6) 20 times in Python and compare.
- Make a family lucky draw. Write names in a Python list and let random.choice() pick who chooses dinner.
- Ask your child to guess what the program will print before running it. Then run it together.
- Encourage small changes, like changing the dice from 6 sides to 10 sides, and see what happens.

🎯 Quick Check (tap to reveal)

1. **1. What line must you write before using the random module?**
   ✅ import random
2. **2. Can random.randint(1, 6) give the number 6?**
   ✅ Yes. Both 1 and 6 can come up.
3. **3. Which tool picks one item from a list?**
   ✅ random.choice()
4. **4. What does random.shuffle(my_list) do?**
   ✅ It mixes up the order of the items in my_list itself.
5. **5. What is the use of random.seed(5)?**
   ✅ It makes the random results repeat the same way every time you run the program.